

Clustering and Triangle Motifs

EdgeBasedModels.jl Vignette 08

Simon Frost

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Introduction

Real contact networks exhibit clustering — the tendency for two people who share a mutual contact to also be connected to each other, forming triangles. This property, also called triadic closure, is ubiquitous in social networks but absent from the standard configuration model in the large- N limit.

Clustering has important epidemiological consequences. When transmission can travel around a triangle, one of the three edges is “redundant”: if A infects B and B infects C , the edge from A to C cannot cause a *new* infection. This correlation means that clustering reduces epidemic severity compared to a tree-like network with the same degree distribution.

The edge-based compartmental model framework handles clustering by introducing a bivariate probability generating function $g(x, y)$ where x tracks single-edge stubs and y tracks triangle-edge stubs. This approach follows Volz (2011) and Miller (2009).

In this vignette we show how to:

1. Build clustered network models using ClusteredPGF
2. Compare clustered and unclustered epidemics
3. Explore how increasing clustering reduces R_0 and epidemic severity

Setup

```
using EdgeBasedModels
using ModelingToolkit
using OrdinaryDiffEq
using Symbolics
using Plots
```

Clustered networks

In a clustered network each node has two types of half-edges:

- Single-edge stubs (s): connect to a random partner (tree-like)
- Triangle-edge stubs (t): connect in groups of three to form triangles

The joint degree distribution is encoded by a bivariate PGF:

$$g(x, y) = \sum_{s,t} p_{s,t} x^s y^t$$

where $p_{s,t}$ is the probability that a node has s single-edge stubs and t triangle-edge stubs. The mean single degree is $\langle s \rangle = g_x(1, 1)$ and the mean triangle degree is $\langle t \rangle = g_y(1, 1)$.

The clustering coefficient measures the fraction of connected triples that are closed into triangles:

$$C = \frac{2\langle t \rangle}{2\langle t \rangle + \langle s \rangle}$$

For a Poisson clustered network with mean single-edge degree κ_s and mean triangle-edge degree κ_t :

$$g(x, y) = e^{\kappa_s(x-1) + \kappa_t(y-1)}$$

```

κ_s = 3.0
κ_t = 1.0
cpgf = clustered_poisson_pgf(κ_s, κ_t)
println("Mean single-edge degree: ", mean_single_degree(cpgf))
println("Mean triangle-edge degree: ", mean_triangle_degree(cpgf))
println("Clustering coefficient: ", clustering_coefficient(cpgf))

```

```

Mean single-edge degree: 3.0exp(0.0)
Mean triangle-edge degree: exp(0.0)
Clustering coefficient: 0.4

```

Each triangle-edge stub participates in one triangle with two partners, so the total mean degree (number of distinct neighbours) is $\langle s \rangle + 2\langle t \rangle$.

Building a clustered SIR model

The key insight of the clustered EBCM is that we need two survival probabilities:

- $\theta_2(t)$: probability a single-edge partner has not transmitted infection
- $\theta_3(t)$: probability a triangle-edge partner has not transmitted infection

The susceptible fraction is then $S(t) = g(\theta_2, \theta_3^2)$, where the square on θ_3 accounts for the two triangle partners per triangle stub.

Let us build both an unclustered and a clustered SIR model with comparable mean degrees.

```

@parameters β γ

# Universal anchors: γ=0.25, R₀=2, β derived per scenario (see plan.md)
γ_val = 0.25
R0_target = 2.0
κ_total = 5.0
T_unclustered = R0_target / κ_total
β_val = T_unclustered * γ_val / (1 - T_unclustered)

# Unclustered: all edges are single-stubs, mean degree 5
pgf_unclustered = poisson_pgf(κ_total)
model_unclustered = build_sir(pgf_unclustered, β, γ; form = :expanded)

# Clustered: κ_s=3, κ_t=1 → total mean degree = 3 + 2(1) = 5
cpgf = clustered_poisson_pgf(3.0, 1.0)
model_clustered = build_clustered_sir(cpgf, β, γ)

```

```

EdgeModelSystem(Model clustered_sir:
Equations (8):
    8 standard: see equations(clustered_sir)
Unknowns (8): see unknowns(clustered_sir)
    pop_R(t)
    pop_I(t)
     $\varphi_3_R(t)$ 
     $\varphi_3_I(t)$ 
    ?
Parameters (3): see parameters(clustered_sir)
     $\rho$ 
     $\beta$ 
     $\gamma$ 
Observed (4): see observed(clustered_sir), Dict{Symbol, Any}(: $\theta_3$  =>  $\theta_3(t)$ , : $\varphi_3_I$  =>  $\varphi_3_I(t)$ , :
1 + 3.0(-1 +  $\theta_2(t)$ ) +  $\theta_3(t)^2$ )), :rho_param =>  $\rho$ , :edge_seed_groups => Any[(entry =  $\varphi_2_I(t)$ , t
1 + 3.0(-1 +  $\theta_2(t)$ ) +  $\theta_3(t)^2$ ) - exp(-1 + 3.0(-1 +  $\theta_2(t)$ ) +  $\theta_3(t)^2$ )* $\rho$ ) / exp(0.0)), (entry =  $\varphi_3$ 
1 + 3.0(-1 +  $\theta_2(t)$ ) +  $\theta_3(t)^2$ )*(1 -  $\rho$ )) / exp(0.0))]))

```

The clustered model tracks more state variables due to the separate single-edge and triangle-edge dynamics:

```

println("Unclustered variables: ", keys(model_unclustered.variables))
println("Clustered variables:   ", keys(model_clustered.variables))

```

```

Unclustered variables: [:R, : $\varphi_I$ , :pop_I, :pop_R, : $\varphi_R$ , : $\theta$ ]
Clustered variables:  [: $\theta_3$ , : $\varphi_3_I$ , : $\varphi_3_R$ , : $\varphi_2_I$ , :pop_I, :R, :pop_R, : $\varphi_2_R$ , : $\theta_2$ ]

```

Comparing clustered vs unclustered epidemics

We now solve both models with the unclustered Poisson baseline anchored at $R_0 = 2$ ($\beta = 1/6$, $\gamma = 0.25$), and compare the epidemic curves.

```

tspan = (0.0, 40.0)
params = Dict( $\beta$  =>  $\beta_{val}$ ,  $\gamma$  =>  $\gamma_{val}$ )

# Unclustered
ic1 = merge(default_initial_conditions(model_unclustered), params)
prob1 = ODEProblem(model_unclustered.system, ic1, tspan)
sol1 = solve(prob1, Tsit5(); saveat = 0.5)

# Clustered

```

```
ic2 = merge(default_initial_conditions(model_clustered), params)
prob2 = ODEProblem(model_clustered.system, ic2, tspan)
sol2 = solve(prob2, Tsit5(); saveat = 0.5)
```

retcode: Success

Interpolation: 1st order linear

t: 81-element Vector{Float64}:

0.0

0.5

1.0

1.5

2.0

2.5

3.0

3.5

4.0

4.5

⋮

36.0

36.5

37.0

37.5

38.0

38.5

39.0

39.5

40.0

u: 81-element Vector{Vector{Float64}}:

[0.0, 0.001, 0.0, 0.0010000000000000009, 0.0, 0.0010000000000000009, 1.0, 1.0]

[0.000144623672614379, 0.001326107805807947, 0.00014461643846619562, 0.001325973247055298,

[0.00033497795415506396, 0.0017352069162356533, 0.0003349379878670594, 0.00173478462655592

[0.0005828644053464031, 0.0022510324558243327, 0.0005827405493116342, 0.002250055298484182

[0.0009034147604395979, 0.0029034455012692355, 0.0009031116558782131, 0.002901456725756914

[0.0013159697685541298, 0.0037299981287621455, 0.0013153165442927257, 0.003726225967734569

[0.001845104935954855, 0.004777700564948291, 0.0018438057262388148, 0.004770859951768346, 0

[0.0025220405153348123, 0.00610529737504897, 0.0025195880956793057, 0.0060932557069970986,

[0.0033860764081957277, 0.007785467336545977, 0.0033816217012115767, 0.007764739705640867,

[0.004486804708450796, 0.009907869260592578, 0.004478935779073265, 0.009872758398233766, 0.

⋮

[0.7684924194637379, 0.007192733932296931, 0.5103843458746748, 0.0013664247117114415, 0.464

[0.7693445792642608, 0.006452802328367567, 0.5105445393464301, 0.0011974197922827636, 0.464

[0.7701087548816637, 0.005786927653077036, 0.5106847513725135, 0.0010495222828236022, 0.464

```
[0.7707938267003962, 0.005187961768522474, 0.5108074749911291, 0.0009201003717910553, 0.464
[0.7714078772463105, 0.004649395858668374, 0.5109149871101936, 0.0008067432479673713, 0.465
[0.7719581911297007, 0.004165360386799442, 0.5110093483369407, 0.0007072612939204892, 0.465
[0.7724512024872386, 0.003730549318561273, 0.5110921803987606, 0.0006199402285695773, 0.465
[0.7728926785078177, 0.0033401151108894125, 0.511164784159956, 0.0005434150928230247, 0.465
[0.7732878931740367, 0.0029896659767265237, 0.5112284260589087, 0.00047634766958477864, 0.4
```

Extract the epidemic compartments using the PGFs:

```
S1 = compartment(sol1, model_unclustered, :S)
I1 = compartment(sol1, model_unclustered, :I)
R1 = compartment(sol1, model_unclustered, :R)

S2 = compartment(sol2, model_clustered, :S)
I2 = compartment(sol2, model_clustered, :I)
R2 = compartment(sol2, model_clustered, :R)
```

81-element Vector{Float64}:

```
0.0
0.000144623672614379
0.00033497795415506396
0.0005828644053464031
0.0009034147604395979
0.0013159697685541298
0.001845104935954855
0.0025220405153348123
0.0033860764081957277
0.004486804708450796
0.7684924194637379
0.7693445792642608
0.7701087548816637
0.7707938267003962
0.7714078772463105
0.7719581911297007
0.7724512024872386
0.7728926785078177
0.7732878931740367
```

```
plot(sol1.t, S1, label = "S (unclustered)", linewidth = 2, color = :blue)
plot!(sol1.t, I1, label = "I (unclustered)", linewidth = 2, color = :red)
```

```

plot!(sol1.t, R1, label = "R (unclustered)", linewidth = 2, color = :green)
plot!(sol2.t, S2, label = "S (clustered)", linewidth = 2, color = :blue, linestyle = :dash)
plot!(sol2.t, I2, label = "I (clustered)", linewidth = 2, color = :red, linestyle = :dash)
plot!(sol2.t, R2, label = "R (clustered)", linewidth = 2, color = :green, linestyle = :dash)
xlabel!("Time")
ylabel!("Fraction of population")
title!("Clustered vs Unclustered SIR ( $R_0=2$  unclustered baseline)")

```

Clustered vs Unclustered SIR ($R_0=2$ unclustered baseline)

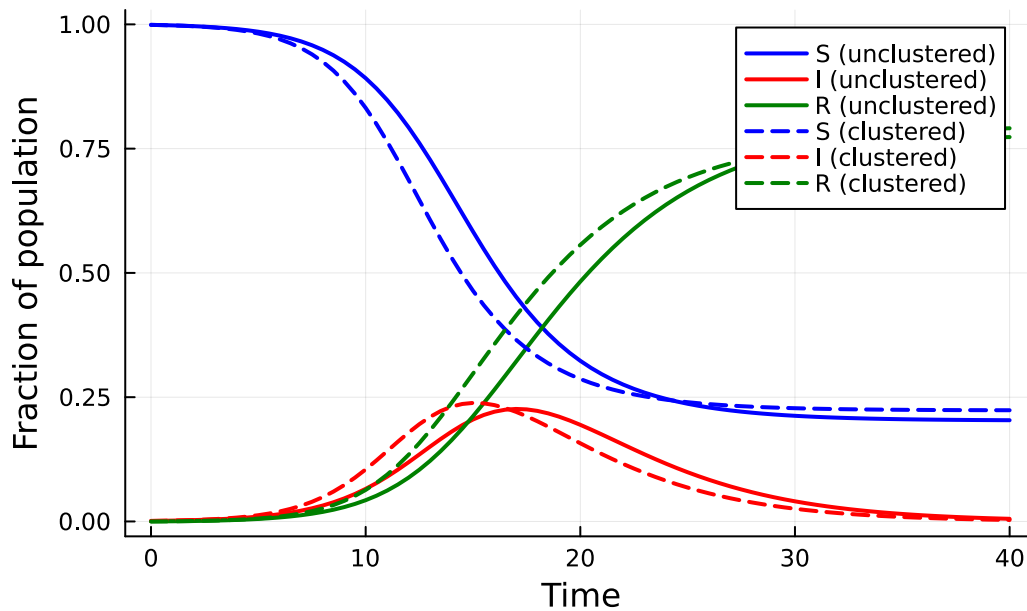


Figure 1: Clustering reduces peak prevalence and delays the epidemic peak. Both networks have mean degree 5.

The dashed (clustered) epidemic has a lower peak prevalence and a delayed peak compared to the solid (unclustered) curves. The final epidemic size is also smaller with clustering.

Varying the clustering coefficient

We fix the total mean degree at 5 and vary the fraction of edges that come from triangles. As clustering increases, more edges are “wasted” on redundant triangle paths.

```

fracs = [0.0, 0.2, 0.4, 0.6]
results = []

for frac in fracs
    κ_s_i = 5.0 * (1 - frac)
    κ_t_i = 5.0 * frac / 2 # each triangle stub contributes 2 neighbours
    cpgf_i = clustered_poisson_pgf(κ_s_i, κ_t_i)
    model_i = build_clustered_sir(cpgf_i, β, γ)
    ic_i = merge(default_initial_conditions(model_i), params)
    prob_i = ODEProblem(model_i.system, ic_i, tspan)
    sol_i = solve(prob_i, Tsit5(); saveat = 0.5)

    S_i = compartment(sol_i, model_i, :S)
    I_i = compartment(sol_i, model_i, :I)
    R_i_vals = compartment(sol_i, model_i, :R)
    C_i = 2κ_t_i / (2κ_t_i + κ_s_i)

    push!(results, (frac = frac, C = C_i, t = sol_i.t, I = I_i, S = S_i, R = R_i_vals))
end

p = plot(xlabel = "Time", ylabel = "Fraction infected",
         title = "Effect of clustering on epidemic dynamics")
colors = [:red, :orange, :purple, :blue]
for (i, res) in enumerate(results)
    plot!(p, res.t, res.I,
          label = "C = $(round(res.C; digits=2)) (frac = $(res.frac))",
          linewidth = 2, color = colors[i])
end
p

```

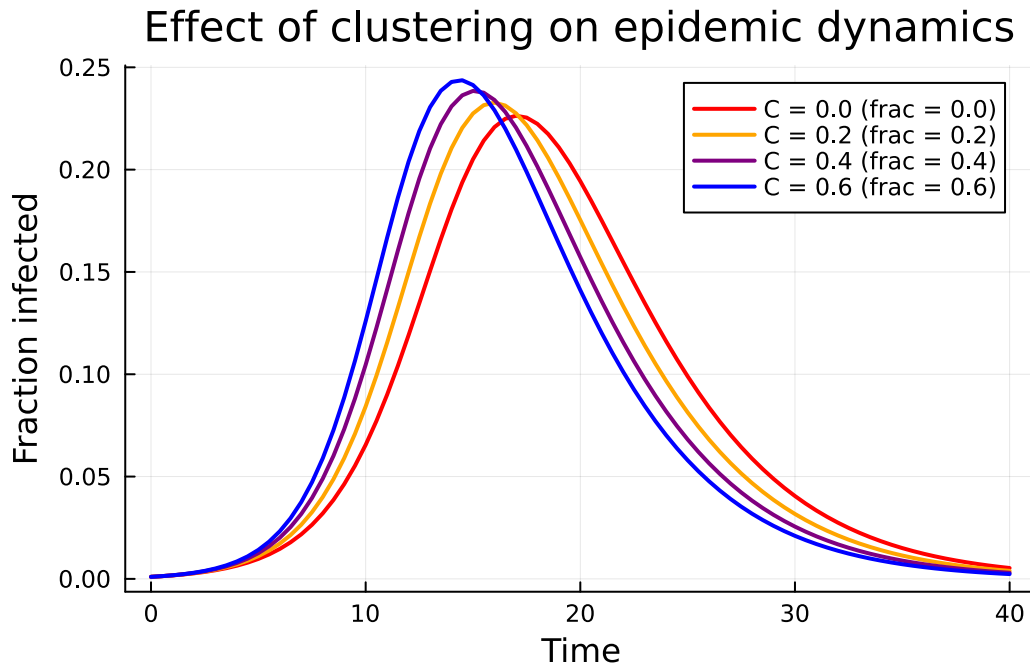



Figure 2: Increasing the fraction of triangle edges reduces epidemic severity while keeping total mean degree = 5.

As the clustering coefficient C increases, the peak prevalence decreases and the epidemic is delayed.

Clustering coefficient and R_0

The basic reproduction number R_0 for a clustered network accounts for the redundant transmission paths within triangles. We can compute it using `basic_reproduction_number`:

```
using EdgeBasedModels: ClusteredConfigurationModel, DiseaseProgression, DiseaseStage, Dise

fracs_fine = 0.0:0.05:0.8
C_vals = Float64[]
R0_vals = Float64[]

for frac in fracs_fine
    κ_s_i = 5.0 * (1 - frac)
    κ_t_i = 5.0 * frac / 2
    cpgf_i = clustered_poisson_pgf(κ_s_i, κ_t_i)
    C_i = 2κ_t_i / (2κ_t_i + κ_s_i)
```

```

push!(C_vals, C_i)

prog = DiseaseProgression(
    [DiseaseStage(:I; transmission_rate =  $\beta$ _val),
     DiseaseStage(:R; transmission_rate = 0)],
    [DiseaseTransition(:I, :R,  $\gamma$ _val)]; entry = :I)
cm = ClusteredConfigurationModel(cpgf_i, prog)
R0_i = basic_reproduction_number(cm)
push!(R0_vals, Float64(Symbolics.value(R0_i)))
end

plot(C_vals, R0_vals, linewidth = 2, color = :darkred, marker = :circle, markersize = 3,
     label = "R0")
xlabel!("Clustering coefficient C")
ylabel!("R0")
title!("R0 vs Clustering (total mean degree = 5, anchored  $\beta$ ,  $\gamma$ )")
hline!([1.0], linestyle = :dash, color = :gray, label = "R0 = 1")

```

R_0 vs Clustering (total mean degree = 5, anchored

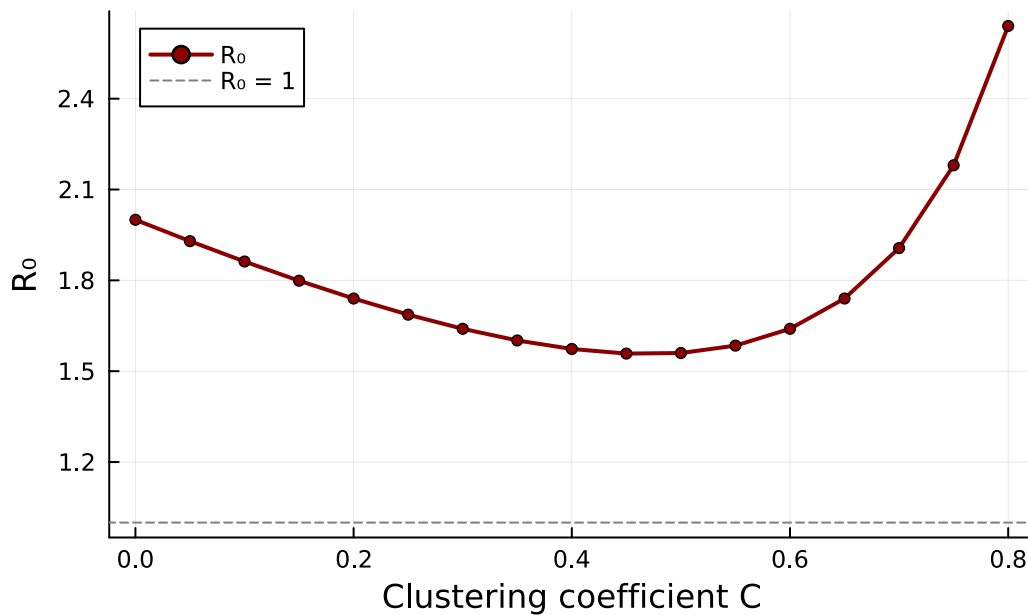


Figure 3: R_0 decreases as the clustering coefficient increases, reflecting the redundancy of triangle edges.

As the clustering coefficient rises from 0 toward 1, R_0 decreases. This quantifies the protective

effect of clustering: triangles create correlated transmission paths that reduce the effective number of secondary infections.

SEIR with clustering

The clustered EBCM framework extends to arbitrary disease progressions. Here we add a latent (exposed) period using `build_clustered_seir`:

```
@parameters  $\sigma$ 
```

```
cp_gf_seir = clustered_poisson_pgf(3.0, 1.0)
model_seir = build_clustered_seir(cp_gf_seir,  $\sigma$ ,  $\beta$ ,  $\gamma$ )
println("SEIR clustered variables: ", keys(model_seir.variables))
```

SEIR clustered variables: [: θ_3 , : ϕ_3_I , : ϕ_3_R , : ϕ_2_I , :pop_I, :R, : θ_2 , : ϕ_2_E , : ϕ_3_E , :pop_R, : ϕ_2_I]

```
params_seir = Dict( $\beta \Rightarrow \beta_{\text{val}}$ ,  $\gamma \Rightarrow \gamma_{\text{val}}$ ,  $\sigma \Rightarrow 0.2$ )
ic_seir = merge(default_initial_conditions(model_seir), params_seir)
prob_seir = ODEProblem(model_seir.system, ic_seir, tspan)
sol_seir = solve(prob_seir, Tsit5(); saveat = 0.5)

S_seir = compartment(sol_seir, model_seir, :S)
R_seir_vals = compartment(sol_seir, model_seir, :R)
I_seir = 1.0 .- S_seir .- R_seir_vals
```

81-element Vector{Float64}:

```
0.00100000000000000009
0.0010130454905676778
0.00104717512584005
0.0010968014650383033
0.0011582546048606796
0.0012291684860312529
0.0013080987956067896
0.0013942005315523895
0.0014870910042569122
0.0015866444132882913
⋮
0.07593289690599012
0.07978103926137754
0.08374450570741221
```

```
0.08781939417069416
0.09200168706017353
0.09628717125938369
0.10066838862572111
0.10513033548489154
0.10966182705534845
```

```
plot(sol_seir.t, S_seir, label = "S", linewidth = 2, color = :blue)
plot!(sol_seir.t, I_seir, label = "E + I", linewidth = 2, color = :red)
plot!(sol_seir.t, R_seir_vals, label = "R", linewidth = 2, color = :green)
xlabel!("Time")
ylabel!("Fraction of population")
title!("Clustered SEIR ( $\kappa_s=3$ ,  $\kappa_t=1$ ,  $\sigma=0.2$ , anchored  $\beta$ ,  $\gamma$ )")
```

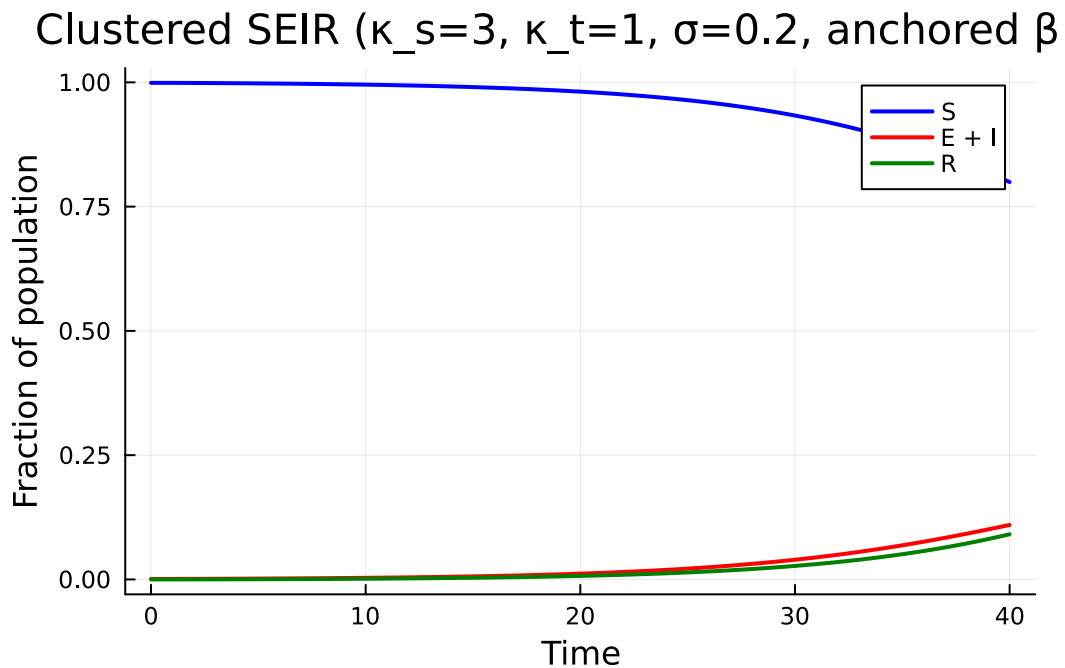


Figure 4: SEIR epidemic on a clustered network with a latent period ($\sigma=0.2$).

The latent period further delays and flattens the epidemic curve on top of the clustering effect.

Summary

- Clustering (triadic closure) is a key structural property of real contact networks that is absent from the standard configuration model.

- Triangles create redundant transmission paths: if two of three triangle edges transmit, the third cannot cause a new infection.
- The clustered EBCM uses a bivariate PGF $g(x, y)$ and tracks separate survival probabilities for single-edge (θ_2) and triangle-edge (θ_3) partners.
- Increasing clustering reduces R_0 and lowers epidemic peak prevalence while keeping the mean degree fixed.
- The `EdgeBasedModels.jl` package provides `clustered_poisson_pgf`, `build_clustered_sir`, and `build_clustered_seir` for modelling clustered networks with arbitrary disease progressions.

Simulation validation

We validate both the unclustered and clustered EBCMs against direct SSA on graphs sampled from the matching configuration models. The unclustered baseline uses an Erdős–Rényi graph at mean degree $\kappa = 5$. The clustered network uses the Newman–Miller doubly Poisson construction: each node draws $\text{Poisson}(\kappa_s = 3)$ single-edge stubs and $\text{Poisson}(\kappa_t = 1)$ triangle-corner stubs, single stubs are randomly paired, and triangle stubs are grouped in threes (3 edges per triangle) — yielding total mean degree $\kappa_s + 2\kappa_t = 5$.

```
include("../_validation.jl")

prog_clust = DiseaseProgression(
    [DiseaseStage(:I; transmission_rate =  $\beta_{\text{val}}$ ), DiseaseStage(:R)],
    [DiseaseTransition(:I, :R,  $\gamma_{\text{val}}$ )]; entry = :I)

# Unclustered SSA ribbon
tg_un, mean_un, std_un = gillespie_ribbon(
    prog_clust, Dict{: $\beta$   $\Rightarrow$   $\beta_{\text{val}}$ , : $\gamma$   $\Rightarrow$   $\gamma_{\text{val}}$ },
    poisson_graph_builder(2000, 5.0);
    N = 2000, n_graphs = 3, nsims_per_graph = 12,
    tspan = tspan, seed_fraction = 0.01,
    tgrid = collect(0.0:0.5:40.0))

# Clustered SSA ribbon
tg_cl, mean_cl, std_cl = gillespie_ribbon(
    prog_clust, Dict{: $\beta$   $\Rightarrow$   $\beta_{\text{val}}$ , : $\gamma$   $\Rightarrow$   $\gamma_{\text{val}}$ },
    clustered_poisson_graph_builder(2000, 3.0, 1.0);
    N = 2000, n_graphs = 3, nsims_per_graph = 12,
    tspan = tspan, seed_fraction = 0.01,
    tgrid = collect(0.0:0.5:40.0))
```

```
([0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5 ... 35.5, 36.0, 36.5, 37.0, 37.5, 38.0, 38.5,
```

```

plot(sol1.t, I1, label = "I (unclustered, EBM)", lw = 2, color = :red)
plot!(tg_un, mean_un[:I], ribbon = std_un[:I],
      label = "I (unclustered, SSA)",
      color = :red, fillalpha = 0.2, linealpha = 0.6, lw = 1)
plot(sol2.t, I2, label = "I (clustered, EBM)", lw = 2, color = :blue)
plot!(tg_cl, mean_cl[:I], ribbon = std_cl[:I],
      label = "I (clustered, SSA)",
      color = :blue, fillalpha = 0.2, linealpha = 0.6, lw = 1)
xlabel!("Time")
ylabel!("Fraction infected")
title!("Clustered vs unclustered SIR: EBM vs SSA")

```

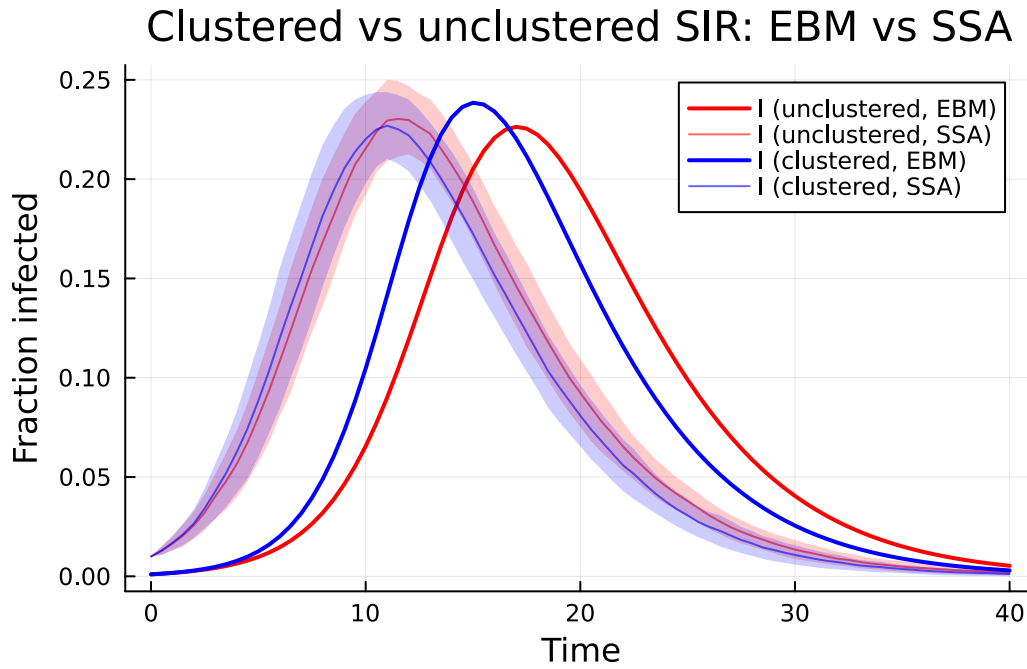


Figure 5: EBM (lines) versus DirectSSA (ribbons = mean $\pm 1\sigma$ across 36 trajectories on 3 graphs of $N=2000$). Both networks have mean degree 5; the clustered network is built via the Newman–Miller doubly Poisson construction with $\kappa_s=3$, $\kappa_t=1$.

The deterministic curves track the SSA ensemble means in both regimes. The clustered network shows the expected reduction in peak prevalence and a delayed/shrunk epidemic, and the SSA on the Newman–Miller graph reproduces this attenuation faithfully — confirming that the bivariate-PGF EBCM correctly captures the redundancy of triangle paths.